

# Algebra and Discrete Mathematics

## ADM\_A\_B

doc. Xiaolu Hou, PhD.

FIIT, STU  
xiaolu.hou @ stuba.sk

# Timetable

- Lectures (prednášky)
  - Friday 9:00 – 11:50
- Tutorials (cvičenia)
  - Friday 12:00 – 13:50
- Consultations (konzultácie)
  - By appointment: [xiaolu.hou@stuba.sk](mailto:xiaolu.hou@stuba.sk)
  - Office 4.14, or online

# Grading

- Semester work – 60 marks
  - Quiz one – 25 marks
    - Covers: Lectures 1–3, Tutorials 1–3
    - Venue: same room as for lecture
    - Week 5
  - Quiz two – 25 marks
    - Covers: Lectures 3, 4, 6, Tutorials 4–7
    - Venue: same room as for lecture
    - Week 9
  - Tutorial submissions – 10 marks
    - There will be 12 tutorial sheets
    - The best 5 submissions count
    - Each submission is worth up to 2 mark
- Final exam – 40 marks
  - To sit the final exam, you must obtain at least 30 marks during the semester.

## Homework

- **Submission deadline:** Tutorials T1–T11: Wednesday, 23:59  
Tutorial T12: Thursday, 23:59
- **Submission method:** Submit via UIS or in person. If you do not have access to UIS and cannot submit in person, you may email me.
- **What counts:** There will be 12 tutorial sheets; your best 5 submissions count toward your final mark.
- **Requirements:** Your solution should be clearly written, logically organised, and include the main steps of the argument, not just the final answer.
- **Marking:** Each submission is graded on a scale 0/1/2:
  - 2: all required parts are meaningfully attempted, and the solutions are complete, clear, and mathematically sound
  - 1: all required parts are meaningfully attempted, but some solutions are incomplete, unclear, or contain mathematical errors
  - 0: one or more required parts are not meaningfully attempted, or there is no meaningful submission.

A meaningful attempt must show relevant reasoning, setup, calculations, or argument; a final answer alone does not count as a meaningful attempt.

- **Late submissions:** Late submissions are not accepted.

# Attendance Policy

- Attendance at tutorials is compulsory.
  - Absences must be justified by a valid medical certificate (MC).
  - In the case of more than two absences without a valid MC:
    - You must submit solutions to **ALL** questions from the missed tutorial within one week.
    - Submissions can be made via email (xiaolu.hou@stuba.sk) or in person.
- Some lessons might be online, pay attention to Teams announcement
- If you do not see the Teams group, please let me know

# Tutorials

- A summary of certain concepts from the lecture
- Discussion of tutorial questions
  - Recommendation: try to solve the questions on your own before the tutorial, then compare your work with the solutions.

# Course Outline

- Vectors and matrices
- System of linear equations
- Matrix inverse and determinants
- Vector spaces, bases and matrix transformations
- Quiz one
- Matrix transformations, fundamental spaces and decompositions
- Eulerian tours and Hamiltonian cycles
- Paths and spanning trees
- Quiz two
- Trees and networks
- Matching

# Textbooks

- Andrilli, Stephen, and David Hecker. *Elementary Linear Algebra*. 5th ed. Academic Press, 2022.
  - [Accessible online \(free copy\)](#)
  - [Alternative download link](#)
- Anton, Howard, and Chris Rorres. *Elementary Linear Algebra: Applications Version*. John Wiley & Sons, 2013.
  - [Accessible online \(free copy\)](#)
  - [Alternative download link](#)
- Saoub, K. R. *A Tour Through Graph Theory*. Chapman and Hall/CRC, 2017.
  - [Accessible online \(free copy\)](#)
  - [Alternative download link](#)

# Course materials

[https://xiaoluhou.github.io/Teaching\\_material/](https://xiaoluhou.github.io/Teaching_material/)

or

[https://github.com/XIAOLUHOU/Teaching\\_material](https://github.com/XIAOLUHOU/Teaching_material)

## Remarks

- **Proofs:** Included for reference and deeper understanding.
- **Errors:** Please report any mistakes by email.